1. 1D slab $x \in [0, a]$ w/ anisotropic scattering & isotropic source $q(x)$
Write down P_N eq & P_n approx to ψ for a) $N=3$ b) $N=5$

a) $N=3$

w/ $\psi^N(x,\mu) = \sum_{\ell=0}^{N} \frac{2\ell+1}{4\pi} \phi_\ell(x) P_\ell(\mu)$

$$\psi^3(x,\mu) = \frac{1}{4\pi}\left[\phi_0(x) + 3\mu\,\phi_1(x) + \frac{5}{2}(3\mu^2-1)\,\phi_2(x) + \frac{7}{2}(5\mu^3-3\mu)\,\phi_3(x)\right]$$

w/ the P_N equations, we obtain the following equations

$\ell=0$: $\quad \frac{d\phi_1}{dx} + (\Sigma_{t0} - \Sigma_{s0})\phi_0 = q(x)$ (assume $q(x)$ is a group $\ell=0$ source)

$\ell=1$: $\quad \frac{2}{3}\frac{d\phi_2}{dx} + \frac{1}{3}\frac{d\phi_0}{dx} + (\Sigma_t - \Sigma_{s1})\phi_1 = 0$

$\ell=2$: $\quad \frac{3}{5}\frac{d\phi_3}{dx} + \frac{2}{5}\frac{d\phi_1}{dx} + (\Sigma_t - \Sigma_{s2})\phi_2 = 0$

$\ell=3$: $\quad \frac{3}{7}\frac{d\phi_2}{dx} + (\Sigma_t - \Sigma_{s3})\phi_3 = 0$

use vacuum conditions: $\frac{d\psi_4}{dx} = 0$

16) $N=5$

$$\psi^N(x,\mu) = \sum_{\ell=0}^{N} \frac{2\ell+1}{4\pi} \phi_\ell(x) P_\ell(\mu)$$

$$\psi^5(x,\mu) = \frac{1}{4\pi}\left[\phi_0(x) + 3\mu\phi_1(x) + \frac{5}{2}(3\mu^2-1)\phi_2(x) + \frac{7}{2}(5\mu^3-3\mu)\phi_3(x) + \frac{9}{8}(35\mu^4-30\mu^2+3)\phi_4(x) + \frac{11}{8}(63\mu^5-70\mu^3+15\mu)\phi_5(x)\right]$$

w/ the corresponding PN equations

$\ell=0$: $\dfrac{d\phi_1}{dx} + (\Sigma_t - \Sigma_{s0})\phi_0 = q(x)$

$\ell=1$: $\dfrac{2}{3}\dfrac{d\phi_2}{dx} + \dfrac{1}{3}\dfrac{d\phi_0}{dx} + (\Sigma_t - \Sigma_{s1})\phi_1 = 0$

$\ell=2$: $\dfrac{3}{5}\dfrac{d\phi_3}{dx} + \dfrac{2}{5}\dfrac{d\phi_1}{dx} + (\Sigma_t - \Sigma_{s2})\phi_2 = 0$

$\ell=3$: $\dfrac{4}{7}\dfrac{d\phi_4}{dx} + \dfrac{3}{7}\dfrac{d\phi_2}{dx} + (\Sigma_t - \Sigma_{s3})\phi_3 = 0$

$\ell=4$: $\dfrac{5}{9}\dfrac{d\phi_5}{dx} + \dfrac{4}{9}\dfrac{d\phi_3}{dx} + (\Sigma_t - \Sigma_{s4})\phi_4 = 0$

$\ell=5$: $\dfrac{5}{11}\dfrac{d\phi_4}{dx} + (\Sigma_t - \Sigma_{s5})\phi_5 = 0$

use closure $\dfrac{d\psi^6}{dx} = 0$

2) for $x=0$, have BC on flux when $x=0, \mu>0$ have following

$$2\pi \int_0^1 \psi^N(0,\mu)\, P_\ell(\mu)\, d\mu = 0 \qquad \ell = 1,3$$

we obtain left BC's by solving eq above for P_1, P_3

for $2\pi \int_0^1 \psi^3 P_1 = 0$

$$\Rightarrow \int_0^1 \frac{1}{4\pi}\left[\phi_0 + 3P_1\phi_1 + \frac{5}{2}P_2\phi_2 + \frac{7}{2}P_3\phi_3\right] P_1 = 0$$

$$\int_0^1 \phi_0 P_1 + 3P_1^2\phi_1 + \frac{5}{2}P_1P_2\phi_2 + \frac{7}{2}P_3P_1\phi_3 = 0$$

break apart integrals

$$\Rightarrow \int_0^1 P_1\, d\mu = \int_0^1 \mu\, d\mu = 1/2$$

$$\Rightarrow \int_0^1 P_1^2\, d\mu = \int_0^1 \mu^2\, d\mu = 1/3$$

$$\Rightarrow \int_0^1 P_1P_2\, d\mu = \int_0^1 \mu\left(\tfrac{1}{2}(3\mu^2-1)\right) d\mu = 1/8$$

$$\Rightarrow \int_0^1 P_1P_3\, d\mu = \int_0^1 \tfrac{1}{2}\mu(5\mu^2-3\mu)\, d\mu = 0$$

combine $\Rightarrow \frac{1}{2}\phi_0 + (3)(\frac{1}{3})\phi_1 + (5)(\frac{1}{8})\phi_2 = 0$

left w/ @ $x=0$ $\quad 4\phi_0(x=0) + 8\phi_1 + 5\phi_2 = 0$

for $2\pi \int_0^1 \mathcal{T}^3 P_3 = 0$

$$\Rightarrow \int_0^1 \frac{1}{4\pi}\left[\Phi_0 + 3P_1\Phi_1 + \frac{5}{2}P_2\Phi_2 + \frac{7}{2}P_3\Phi_3\right] P_3\, d\mu = 0$$

$$\int_0^1 P_3\Phi_0 + 3P_1P_3\Phi_1 + \frac{5}{2}P_2P_3\Phi_3 + \frac{7}{2}P_3^2\Phi_3 \; d\mu = 0$$

breaks apart integrals

$$\Rightarrow \int_0^1 P_3\, d\mu = \int_0^1 \tfrac{1}{2}(5x^3-3x)\, d\mu = \tfrac{-1}{8}$$

$$\Rightarrow \int_0^1 P_1P_3\, d\mu = \int_0^1 \tfrac{1}{2}(5x^3-3x)\mu\, d\mu = 0$$

$$\Rightarrow \int_0^1 P_2P_3\, d\mu = \int_0^1 \tfrac{1}{4}(5x^3-3x)(3x^2-1)\, d\mu = \tfrac{1}{8}$$

$$\Rightarrow \int_0^1 P_3^2\, d\mu = \int_0^1 \tfrac{1}{4}(5x^3-3x)\, d\mu = \tfrac{1}{7}$$

combine $\Rightarrow \left(\frac{-1}{8}\right)\Phi_0 + \left(\frac{1}{8}\right)(5)\Phi_2 + \left(\frac{1}{7}\right)(7)P_3 = 0$

left w/ @ $x=0$ $\quad -\Phi_0 + 5\Phi_2 + 8\mathcal{T}_3 = 0$

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odd legendre ~~polynomials~~ are odd s.t. $P_{odd}(-\mu) = -P_{odd}(\mu)$
& even legendre are even s.t. $P_{even}(\mu) = P_{even}(-\mu)$

∴ we can do the same for $x=a$ w/ $2\pi \int_{-1}^{0} \mathcal{T}^N(a,\mu) P_2(\mu)\, d\mu = 0$, we obtain similar expressions for $x=a$

- odd terms flip signs

side	P_1 evaluated	P_3 evaluated
$x=0$, Left	$4\Phi_0(0) + 8\Phi_1(0) + 5\Phi_2(0) = 0$	$-\Phi_0(0) + 5\Phi_2(0) + 8\Phi_3(0) = 0$
$x=a$, Right	$4\Phi_0(a) - 8\Phi_1(a) + 5\Phi_2(a) = 0$	$-\Phi_0(a) + 5\Phi_2(a) - 8\Phi_3(a) = 0$

3) Start w/ DPN expansion

$$\psi(x,\mu) = \sum_{\ell=0}^{N} \frac{2\ell+1}{4\pi}\left(\phi_\ell^+(x) P_\ell^+(\mu) + \phi_\ell^-(x) P_\ell^-(x)\right)$$

w/ $\phi_\ell^+ = 2\pi \int_0^1 \psi(x,\mu) P_\ell^+(\mu) d\mu$ $\qquad$ $\phi_\ell^- = 2\pi \int_{-1}^0 \psi(x,\mu) P_\ell^-(\mu) d\mu$

$P_\ell^+(\mu) = P_\ell(2\mu-1)$ for $\mu>0$ $\qquad$ $P_\ell^-(\mu) = P_\ell(2\mu+1)$ for $\mu<0$

For DP_3, have following

- for $\mu>0$: $\psi(x,\mu) = \frac{1}{4\pi}\left[\phi_0^+(x) + 3(2\mu-1)\phi_1^+(x)\right]$
- for $\mu<0$: $\psi(x,\mu) = \frac{1}{4\pi}\left[\phi_0^-(x) + 3(2\mu+1)\phi_1^-(x)\right]$

we plug these two one equations into the DPN equations

- for $\mu>0$:

  - $\ell=0$: $\frac{d\phi_1^+}{dx} + 2\Sigma_t \phi_0^+ = \Sigma_s(\phi_0^+ + \phi_0^-) + 2q$

  - $\ell=1$: $\frac{d\phi_0^+}{dx} + 3\Sigma_t \phi_1^+ = 0$

For $\mu<0$:

- $\ell=0$: $-\frac{d\phi_1^-}{dx} + 2\Sigma_t \phi_0^- = \Sigma_s(\phi_0^+ + \phi_0^-) + 2q$

- $\ell=1$: $\frac{d\phi_0^-}{dx} + 3\Sigma_t \phi_1^- = 0$

3) Vacuum BC @ $x=0$

- BC means $\psi(0,\mu)=0$ for $\mu>0$

- we only have BC on positive μ, so only work from $\int_0^1 P_\ell^+ d\mu$

$$\int_0^1 \psi(0,\mu) P_\ell^+(\mu)\, d\mu = \int_0^1 (0) P_\ell^+(\mu)\, d\mu = 0$$

expand ψ: $\int_0^1 \frac{1}{4\pi}\left[\sum_{\ell=0}^{L} (2\ell+1)\phi_\ell^+(0) P_\ell^+(\mu)\right] P_k^+(\mu)\, d\mu = 0$ (P_k)

$$\Rightarrow \frac{1}{4\pi}\sum_{\ell=0}^{L}(2\ell+1)\phi_\ell^+(0)\int_0^1 P_\ell^+(\mu) P_k^+(\mu)\, d\mu = 0$$

w/ $\int_0^1 P_\ell^+(\mu) P_k^+(\mu)\, d\mu = \frac{1}{2\ell+1}\delta_{\ell k}$ from definition

$$\Rightarrow \sum_{\ell=0}^{L}\phi_\ell^+(0) = 0 \qquad \therefore\ \phi_0^+(0)=0\ ,\ \phi_1^+(0)=0 \text{ for } \mu>0$$

Reflecting BC @ $x=a$

- Reflecting means $\psi(a,-\mu) = \psi(a,\mu)$ for $\mu>0$

- substitute expressions from before w/ DPN expansion

$$\frac{1}{4\pi}\left[\phi_0^-(a)P_0^-(-\mu) + 3\phi_1^-(a)P_1^-(-\mu)\right] = \frac{1}{4\pi}\left[\phi_0^+(a)P_0^+(\mu) + 3\phi_1^+(a)P_1^+(\mu)\right]$$

- use fact $P_{even}^-(-\mu) = P_{even}^+(\mu)$, $P_{odd}^-(-\mu) = -P_{odd}^+(+\mu)$

$$\Rightarrow \phi_0^-(a)P_0^+(\mu) - 3\phi_1^-(a)P_1^+(\mu) = \phi_0^+(a)P_0^+(\mu) + 3\phi_1^+(a)P_1^+(\mu) \quad (11)$$

3) • multiply (11) eq $P_0^+(\mu)\,d\mu$ & integrate over $\int_0^1$

$$\int_0^1 \left[P_0^{+2}\phi_0^-(a) - 3P_0^+P_1^+\phi_1^-(a)\right]d\mu = \int_0^1 \left[P_0^{+2}\phi_0^+(a) + 3P_0^+P_1^+\phi_1^+\right]d\mu$$

• P_0^+ & P_1^+ are orthogonal over $[0,1]$, so $\int_0^1 P_0^+P_1^+\,d\mu = 0$

$$\Rightarrow \phi_0^-(a)\int_0^1 [P_0^+(\mu)]^2\,d\mu = \phi_0^+(a)\int_0^1 [P_0^+(\mu)]^2\,d\mu$$

$$\therefore \underline{\phi_0^-(a) = \phi_0^+(a)}$$

• Multiply (11) eq $P_1^+(\mu)\,d\mu$ & integrate over $\int_0^1$

$$\int_0^1 \left[P_0^+P_1^+\phi_0^-(a) - 3P_1^{+2}\phi_1^-(a)\right]d\mu = \int_0^1 \left[P_0^+P_1^+\phi_0^+(a) + 3P_1^{+2}\phi_1^+(a)\right]d\mu$$

• once again, apply orthogonality condition $\int_0^1 P_0^+P_1^+\,d\mu = 0$

$$\Rightarrow 3\phi_1^-(a)\int_0^1 P_1^{+2}(\mu)\,d\mu = -3\phi_1^+(a)\int_0^1 P_1^{+2}(\mu)\,d\mu$$

$$\therefore \underline{-\phi_1^-(a) = \phi_1^+(a)}$$

1st term

4) Start by evaluating ϕ w/ P_1 angular flux

$$\phi = \int_{4\pi} \left[\phi_{0,0} Y_{0,0} + \sum_{m=-1}^{1} \phi_{1,m} Y_{1,m} \right] d\vec{\Omega}$$

recall $Y_{(0,0)}(\theta, \gamma) = \sqrt{1/4\pi}$, which is constant. we can say any spherical harmonic is integrated by a multiple of $Y_{0,0}$, so, the integrals of all higher order moments equal 0.

$$\Rightarrow \phi = \phi_{0,0} \int_{4\pi} Y_{0,0} \, d\vec{\Omega} = \phi_{0,0} \left(\frac{1}{\sqrt{4\pi}} \cdot 4\pi \right) = 2\sqrt{\pi}\, \phi_{0,0} = \sqrt{4\pi}\, \phi_{0,0}$$

$$\Rightarrow \phi_{0,0} = \phi / \sqrt{4\pi} \quad \& \quad \underline{\phi_{0,0} Y_{0,0} = \frac{1}{4\pi} \phi(r)}$$

Current terms

I will choose a cartesian $\hat{\Omega}$ such that

$$\vec{\Omega} = \Omega_x \hat{x} + \Omega_y \hat{y} + \Omega_z \hat{z} = \sin\theta \cos\gamma \, \hat{x} + \sin\theta \sin\gamma \, \hat{y} + \cos\theta \, \hat{z}$$

next, evaluate the following spherical harmonics

- $Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos\theta = \sqrt{\frac{3}{4\pi}} \Omega_z$
- $Y_{1,-1} = \sqrt{3/8\pi} \sin\theta \exp(-i\gamma) = \sqrt{3/8\pi} \sin\theta (\cos\gamma - i \sin\gamma) = \sqrt{3/8\pi} (\Omega_x - i\Omega_y)$
- $Y_{1,1} = \sqrt{3/8\pi} \sin\theta \exp(i\gamma) = \sqrt{3/8\pi} \sin\theta (\cos\gamma + i\sin\gamma) = \sqrt{3/8\pi} (\Omega_x + i\Omega_y)$

4) • we can add then subtract the third equation from the second

$$Y_{1,1} + Y_{1,-1} = -\sqrt{\frac{3}{2\pi}}\, i\,\Omega_y \Rightarrow \Omega_y = i\sqrt{2\pi/3}\,(Y_{1,1} + Y_{1,-1})$$

$$Y_{1,1} - Y_{1,-1} = -\sqrt{3/2\pi}\,\Omega_x \Rightarrow \Omega_x = \sqrt{2\pi/3}\,(Y_{1,-1} - Y_{1,1})$$

& $\Omega_z = \sqrt{4\pi/3}\, Y_{1,0}$

• substitute each into ρ_1 expansion & into current equation

$$\vec{J} = \int_{4\pi} \vec{\Omega}\,\psi\, d\vec{\Omega}$$ from above, $\vec{\Omega}$ was represented as a combination of ~~the~~ three terms $Y_{1,\xi=-1,0,1}$

~~These terms~~ are said vectors which are ~~orthogonal~~ to each other & $Y_{0,0}$. $\therefore Y_{0,0}$ term & $\phi_{0,0} = 0$

• Deal w/ $\vec{J}$ components/terms. note, J_x can only contain $Y_{1,\pm1}$ terms from ψ

$$J_x = \int_{4\pi} \sqrt{\frac{2\pi}{3}}\,(Y_{1,-1} - Y_{1,1})(\phi_{1,-1} Y_{1,-1} + \phi_{1,1} Y_{1,1})\, d\vec{\Omega}$$

$$= \sqrt{\frac{2\pi}{3}}\,(\phi_{1,-1} - \phi_{1,1})$$ as all non-similar terms $(m_1 \neq m_2)$ go to 0

• Similarly for J_y

$$J_y = \int_{4\pi} i\sqrt{\frac{2\pi}{3}}\,(Y_{1,1} + Y_{1,-1})(\phi_{1,-1} Y_{1,-1} + \phi_{1,1} Y_{1,1})\, d\vec{\Omega}$$

$$= i\sqrt{\frac{2\pi}{3}}\,(\phi_{1,1} + \phi_{1,-1})$$

- Finally for $J_z = \int_{4\pi} \left(\sqrt{\frac{4\pi}{3}} Y_{1,0}\right)\left(\phi_{1,0} Y_{1,0}\right) d\vec{\Omega}$

$$J_z = \sqrt{\frac{4\pi}{3}}\,\phi_{1,0}$$

- use $J_x \pm J_y i$ to ~~obtain the~~ following

- $\phi_{1,-1} = \sqrt{3/8\pi}\,(J_x + i J_y)$

- $\phi_{1,1} = -\sqrt{3/8\pi}\,(J_x - i J_y)$

- ~~Recursively~~ we can now find what each of the following equal

$$\phi_{1,0} Y_{1,0} = \left(\sqrt{\frac{3}{4\pi}} J_z\right)\left(\sqrt{\frac{3}{4\pi}} \Omega_z\right) = \frac{3}{4\pi} J_z \Omega_z$$

$$\left(\phi_{1,1} Y_{1,1} + \phi_{1,-1} Y_{1,-1}\right) = \frac{3}{8\pi}(J_x - i J_y)(\Omega_x + i\Omega_y) + \frac{3}{8\pi}(J_x + i J_y)(\Omega_x - i\Omega_y)$$

$$= \frac{3}{4\pi}(J_x \Omega_x + J_y \Omega_y)$$

- all three equations yield the following

$$\sum_{m=-1}^{1} \phi_{1,m} Y_{1,m} = \frac{3}{4\pi}\left(J_x \Omega_x + J_y \Omega_y + J_z \Omega_z\right) = \underline{\frac{3}{4\pi}\vec{J}\cdot\vec{\Omega}}$$

- use $\frac{1}{4\pi}\phi(r) = \phi_{0,0} Y_{0,0}$ from before up ↑ to write

$$\psi(\vec{r},\vec{\Omega}) = \frac{1}{4\pi}\phi(\vec{r}) + \frac{3}{4\pi}\vec{J}(\vec{r})\cdot\vec{\Omega}$$
